

A Novel Approach to OFDM Radar Processing

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Abstract—In this paper a novel approach for the calculation of the radar range profile in OFDM radar systems is presented. The proposed algorithm operates directly on the modulation symbols and overcomes the typical drawbacks of correlation based baseband signal processing. An example system configuration as well as simulation results from a dedicated MatLab model will be presented and discussed.

I. INTRODUCTION

Even though the idea of applying orthogonal frequency division multiplexing (OFDM) signals in radar systems has been discussed since several years, radar systems based on OFDM signals have never become popular. At first view this might be surprising, since there have been discovered many properties of OFDM based radar that are promising interesting features. First, OFDM signals are reported to be advantageous regarding Doppler shift [1]. Besides providing high tolerance against Doppler shift they also do not experience range-Doppler coupling, which allows for independent and unambiguous range and Doppler processing. Second, due to their origin from digital communications, OFDM signals are designed to carry information, which allows for simultaneous information transmission in parallel to radar sensing [2]. This is an interesting feature in particular concerning security applications. However, there are also major drawbacks related to OFDM radar. Since the auto-correlation properties of OFDM signals are not ideal, in the case of classic correlation based processing high sidelobes may occur in the radar image. This can only be circumvented by transmitting fixed code sequences with special properties [3], which does then exclude the option of transmitting information in parallel to radar sensing.

In this paper a novel approach to OFDM radar processing will be presented, that does not involve correlation processing of the baseband signals but directly operates on the complex modulation symbols that compose the OFDM signal. It will be shown that with this approach it will be possible to overcome the drawbacks of the correlation based approach and still the possibility of transmitting arbitrary information in parallel to the radar operation will be given. In the following chapters, first the structure of OFDM signals will be discussed and the proposed novel processing approach will be presented. Then, the parameterization and implementation of a dedicated

OFDM radar system model that has been created with MatLab will be introduced. Finally, simulation results for both the classic correlation based processing and the novel processing approach will be presented, that prove the operability and the superior performance of the new concept.

II. OFDM RADAR PROCESSING

An OFDM signal is generated by multiplexing a series of complex modulation symbols onto multiple orthogonal subcarriers. The OFDM time domain signal of one OFDM symbol can be expressed as

$$x(t) = \sum_{n=0}^{N-1} I(n) \exp(j2\pi f_n t), \quad 0 \leq t \leq T \quad (1)$$

with N denoting the number of orthogonal subcarriers, f_n being the individual subcarrier frequencies, T being the elementary OFDM symbol duration, and $\{I(n)\}$ representing an arbitrary information series consisting of complex modulation symbols obtained through a discrete phase modulation technique, e.g. phase shift keying (PSK). In order to avoid interference between the single subcarriers, the subcarriers have to be orthogonal, which is fulfilled in case of

$$\Delta f = \frac{1}{T} \quad (2)$$

where Δf represents the frequency difference between two adjacent subcarriers. In the frequency domain the spectrum of one OFDM symbol corresponds to

$$X(f) = \sum_{n=0}^{N-1} I(n) \sqrt{T} \frac{\sin(\pi(f - f_n)T)}{(\pi(f - f_n)T)} \quad (3)$$

The OFDM radar platform is supposed to be a monostatic system equipped with one transmitter and one receiver. The radiated signal, which may also simultaneously transmit information to a distant receiver, will be scattered from objects

in the neighborhood of the platform and the receiver that is co-located to the transmitter will receive these scattered waves. The co-located receiver shares the transmitted information $\{I(n)\}$ and can use this information for the radar processing. It is assumed that the radar processing is performed with one single OFDM symbol described by (1) and (3) which is generated from an arbitrary information sequence $\{I(n)\}$ without any specific restrictions. A possible approach to the realization of this idea has already been investigated in [4]. In the cited work it is proposed to calculate the radar range profile by simply correlating the received time domain signal $y(t)$ with the transmitted time domain signal $x(t)$

$$\varphi_{yx}(\tau) = \int y(t)x(t-\tau)dt \quad (4)$$

with τ being the time variable of the radar range profile. This approach is justified with the pulse-like structure of the time domain OFDM signal. However, the proposed approach has two major drawbacks. First, the auto-correlation function of the time domain OFDM signal is not ideal, which results in relatively high sidelobes in the radar range profile. This is a severe drawback, since it drastically limits the dynamic range in scenarios with multiple scattering objects. Second, the correlation properties depend on the transmitted information $\{I(n)\}$, which makes it impossible to guarantee a reliable system performance while transmitting arbitrary information.

In order to circumvent these disadvantages, now a novel approach for OFDM radar processing will be discussed. The basic idea of this approach consists in comparing the transmitted information $\{I(n)\}$ and the received soft-state information $\{I_r(n)\}$ obtained in the receiver at the output of the OFDM de-multiplexer before the channel equalization and the decoding is performed. At this point the distortion from the channel is fully contained in the complex modulation symbols $\{I_r(n)\}$. Since all information symbols in one OFDM symbol are transmitted through the channel at different carrier frequencies separated by Δf , the received information symbols can be used in order to perform channel sensing at discrete frequencies like in stepped frequency radar. The samples of the frequency domain channel transfer function can easily be obtained by simply calculating an element-wise division

$$I_{div}(n) = \frac{I_r(n)}{I(n)}. \quad (5)$$

The sampled channel impulse response, which corresponds to the radar range profile, is obtained as the inverse (discrete) Fourier transform of $\{I_{div}(n)\}$

$$\begin{aligned} h(k) &= \text{IDFT}(\{I_{div}(n)\}) \\ &= \frac{1}{N} \sum_{n=0}^{N-1} I_{div}(n) \exp\left(j \frac{2\pi}{N} nk\right), \quad k = 0, \dots, N-1 \end{aligned} \quad (6)$$

As a consequence of the sampling of the spectrum the resulting radar range profile is periodic in time. However, the radar range profile is not necessarily time-discrete but a continuous function in time. The time-continuous radar range profile can be calculated by a transform that corresponds to the discrete-time Fourier transform

$$h(t) = T \sum_{n=0}^{N-1} I_{div}(n) \exp\left(j \frac{2\pi}{T} nt\right). \quad (7)$$

With this processing approach all drawbacks that occur in the classic baseband processing approach described in (4) will not exist anymore. The dynamic range is only limited by the Fourier transform sidelobes, which do not cause a significant reduction of performance and can even be reduced by applying windowing functions, as usually done e.g. in FMCW radar systems. Also the novel processing approach is completely independent from the transmitted information, since it relates every received modulation symbol to a transmitted one. This fact guarantees a constant and reliable system performance independent from the transmitted information. At the same time the computational effort is even reduced since discrete Fourier operations can be implemented much more efficient than correlations. The only minor disadvantage consists in the fact that a periodicity occurs in the radar range profile, which causes ambiguities if an improper subcarrier spacing is chosen.

III. SYSTEM PARAMETERIZATION AND MODELING

In order to guarantee reliable system performance, several criteria have to be considered when choosing the system parameters. This concerns in particular effects that cause signal distortion like Doppler shift and multi-path propagation. In the following an example system parameterization for operation in the 24 GHz ISM band will be derived. Nevertheless, the described parameterization procedure can be adapted to any suitable band that provides sufficient radar reflectivity and tolerable attenuation for the communications application.

A. Doppler Shift

In comparison to communications propagation in the case of radar propagation we obtain twice the Doppler shift, which results in

$$f_D = \frac{2v_{rel}}{\lambda} \quad (8)$$

with f_D denoting the Doppler frequency shift and v_{rel} being the relative velocity between the system platform and the scatterer. Assuming a maximum relative velocity of $v_{rel} = 200$ km/h corresponding to $v_{rel,max} = 55.6$ m/s this results at the carrier frequency of 24 GHz in a maximum Doppler shift of $f_{D,max} = 8.9$ kHz. In order to not destroy the orthogonality of the subcarriers, the subcarrier spacing Δf has to be chosen much larger than the maximum Doppler shift $f_{D,max}$. It can be assumed that the limit $\Delta f > 10 f_{D,max}$ will

definitely ensure the orthogonality of the subcarriers and even allow for relative velocities distinctly higher than $v_{rel,max}$.

B. Multi-Path Propagation

In order to avoid inter-symbol interference between subsequent OFDM symbols, each OFDM symbol must be extended at the transmitter with a prefix containing a partial cyclic repetition of itself of duration T_p . This duration must correspond to the maximum time difference of the received signal components. For the radar application it must be assumed that there is a direct coupling between the transmitting and the receiving antenna on the same platform. Hence, the cyclic prefix duration must be equal to the total maximum signal travel time between platform and scatterer instead of only the time difference as for pure communication links. Again, for the radar application there is a factor of 2 involved, since the signal travels the distance between platform and scatterer twice. If we assume that in the intended ISM application due to the low transmit power the maximum detectable distance is only 200 m, we obtain $T_p > 1.33 \mu s$. For radar applications that are intended to cover larger distances this parameter can easily be modified.

C. Range Ambiguity

The radar range profile resulting from the proposed novel processing approach is periodic in time with a periodicity equal to the OFDM symbol duration. Considering that the signal travels the distance twice, this results in a maximum unambiguous measurement distance of

$$d_{max} = \frac{c_0}{2\Delta f} = \frac{Tc_0}{2}. \quad (9)$$

Hence, the OFDM symbol duration must be chosen sufficiently large in order to avoid aliasing in the radar range profile. However, this limit is absolutely not critical. Assuming again a maximum detectable distance of 200 m, we obtain $T > 1.33 \mu s$ or $\Delta f < 750 \text{ kHz}$, respectively.

D. Radar Range Resolution

The range resolution does neither depend on the employed waveform nor on the particular parameterization of the OFDM system but only on the total bandwidth occupied by the transmitted signal. The relation is as follows

$$\Delta r = \frac{c_0}{2B} \quad (10)$$

with Δr denoting the range resolution and B being the total signal bandwidth. For the intended application in the 24 GHz ISM band the available bandwidth is around 100 MHz, which corresponds to a range resolution of $\Delta r = 1.5 \text{ m}$.

E. OFDM Symbol Duration

In order to obtain a sufficiently high signal to noise ratio at the input of the radar processor, the energy content of the signal to be processed must be maximized. For practical

applications, where the transmit power is limited, this means the integration time of the processor must be chosen as long as possible. Hence, the OFDM symbol duration must be chosen as long as possible without violating the constraints resulting from the Doppler effect.

F. Number of Subcarriers

In OFDM systems typically powers of 2 are chosen for the number of subcarriers because this allows for efficient implementation of the necessary DFT and IDFT calculations in the transmitter and the receiver. With the already fixed parameters a bandwidth close to 100 MHz is obtained by choosing the number of subcarriers to $N = 1024$.

The final parameter set that will be used for the simulations is shown in Table 1. Some parameter values from the previous considerations have been slightly modified in order to obtain round numbers.

TABLE I.

OFDM Radar System Parameters		
<i>Symbol</i>	<i>Quantity</i>	<i>Value</i>
f_c	Carrier frequency	24 GHz
N	Number of subcarriers	1024
Δf	Subcarrier spacing	90.909 kHz
T	Elementary OFDM symbol duration	11 μs
T_p	Cyclic prefix length	1.375 μs
T_{sym}	Transmit OFDM symbol duration	12.375 μs
B	Total signal bandwidth	93.1 MHz
Δr	Radar range resolution	1.61 m
d_{max}	Unambiguous range	1650 m

G. System Model

All components of the system model are implemented as individual functions in MatLab. The transmitter generates an OFDM signal with the parameters specified in Table 1 from random binary information by applying 4-PSK subcarrier modulation. The sampled time domain signal is then converted into a quasi-continuous signal by oversampling with a sample-and-hold element with subsequent low-pass filtering. Finally, the signal is converted to the carrier frequency in a bandpass sub-sampling representation.

The following stage is a channel model that simulates the scattering of the transmitted signal at an arbitrary number of point scatterers. For each scatterer time delay, phase shift, and attenuation are calculated individually. The total attenuation is the product of free space attenuation and the attenuation caused by the scattering process, which results in

$$a = \sqrt{\frac{\lambda^2 \sigma}{(4\pi)^3 r^4}} \quad (11)$$

with a denoting the signal attenuation in terms of voltage, σ being the radar cross section of the scattering object, and r being the distance to the scattering object. Each scatterer can be assigned an individual relative velocity, which is modeled by converting the signal to the resulting frequency (carrier plus Doppler). Then, the signal components resulting from the different scatterers are superimposed. Since the primary aim is to verify the proposed processing approach, no particular assumptions on transmit power and antenna gain are made.

At the receiver the signal is converted back to the baseband and sampled at the modulation symbol rate. Then a DFT processing is performed in order to recover the modulation symbols in the frequency domain. Finally, the radar processing algorithm is applied.

IV. SIMULATION RESULTS

With the system model simulations have been performed in order to verify the operability of the proposed processing approach and also in order to compare its performance with the classical baseband correlation processing. Both processors have been tested with one single point scatterer with radar cross section $\sigma = 10 \text{ dBm}^2$ and relative velocity $v_{rel} = 0$ at a distance of $d = 30 \text{ m}$. The radar range profile calculated with the classical approach described in (4) is plotted in Fig. 1.

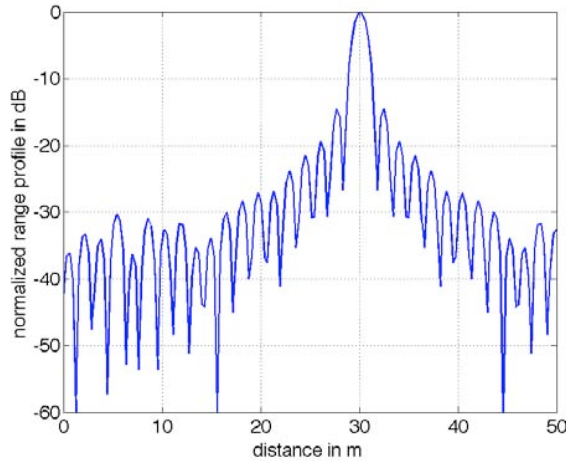


Fig. 1. Radar range profile calculated with the classic correlation based approach for one single scatterer with $\sigma = 10 \text{ dBm}^2$, $v_{rel} = 0$, at $d = 30 \text{ m}$

The scatterer is sharply depicted at a distance of 30 m. The highest sidelobes are at a level of -15 dB, throughout the entire radar image there are sidelobes at a level of at least -30 dB, which look different from those typically resulting from Fourier transforms. These sidelobes are caused by the non-ideal auto-correlation properties of the OFDM signal and cannot be reduced by signal processing techniques like e.g. windowing.

The same scenario has also been processed with the novel OFDM radar processing approach proposed in this paper. The inverse Fourier processing has been performed with an IDFT according to (6) with additional application of a Hamming window. In order to reveal the exact shape of the response from the scatterer, the output series from the IDFT has been

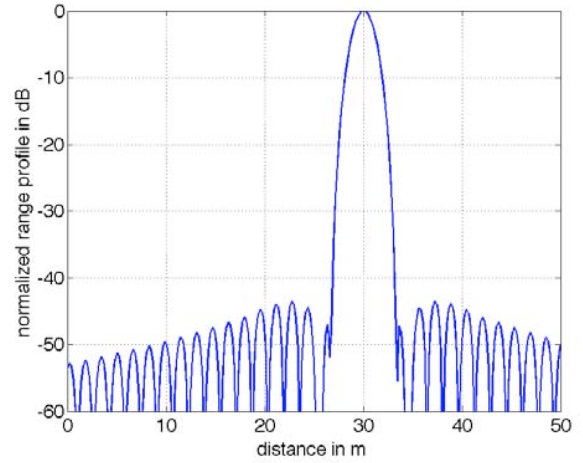


Fig. 2. Radar range profile calculated with the novel approach for one single scatterer with $\sigma = 10 \text{ dBm}^2$, $v_{rel} = 0$, at $d = 30 \text{ m}$

interpolated by a factor of 8. The simulation result is shown in Fig. 2. Again, the scatterer is sharply depicted at a distance of 30 m. The highest sidelobes are at a level of only -43.5 dB and are resulting from the Fourier transform with Hamming window. Sidelobes caused by auto-correlation of the transmitted information do not occur. Hence it has been proven that the proposed approach is working with real OFDM signals. Furthermore, an excellent sidelobe performance is obtained that exceeds the results of the classical baseband correlation processing approach by far.

V. CONCLUSIONS

In this paper a novel concept for the range processing in OFDM radars has been presented. The proposed approach operates directly on the modulation symbols instead of the baseband signals. With this proceeding two major advantages can be achieved. First, the performance of the radar operation is completely independent from the transmitted information. Second, a superior dynamic range compared to the classical correlation based processing is achieved, which is only limited by the sidelobes of the Fourier transform. The novel approach also offers advantages regarding computational effort, since no correlation functions have to be computed. With MatLab simulations the operability of the proposed concept and its superior performance have been proven.

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